

Embedded and IOT Systems



E&ICT Academy, IIT Kanpur
A Joint Initiative of MeitY & IIT Kanpur



Today's Agenda

- **What is Node-RED?**
- **What is flow-based programming?**
- **History of Node-RED**
- **Basic nodes**
- **Installation of Node-RED**

Introduction about NodeRED



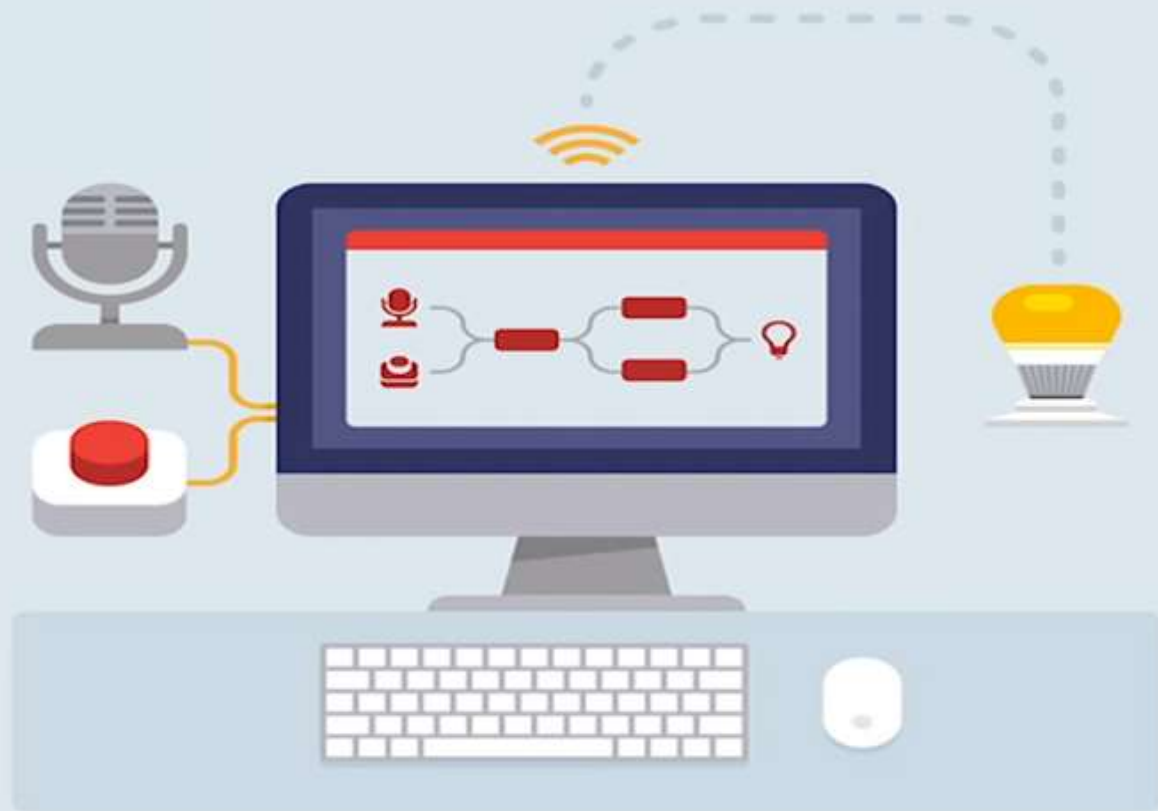
Node-RED

Everything you need to know about Node-RED

- Node-RED is a **programming tool** for **wiring together hardware devices, APIs and online services** in new and interesting ways.
- It provides a **browser-based editor**.
- Since early 2014, the mobile has overtaken the personal computer (desktop/laptop) as the leading device used to navigate the Net.
- Along with the mobile, a number of other portable devices that connect to the Internet have also started proliferating at a very quick rate.
- Nowadays, most of us carry or possess at least one Internet-based device and mobile. So, the Internet of Things (**IoT**) now doesn't only mean different 'things', but has evolved into 'intelligent things' which have onboard computation and network connections. Most importantly, they have the capability to sense the environment around us and, accordingly, act intelligently. These devices are now referred to as connected devices, smart objects or the Web of Things.

Everything you need to know about Node-RED

- This adaptation of technology has led to many developments in the **visual programming** and **visual coding arenas**.
- As a result, it has evolved into a **visual programming/coding** tool based on the already popular Node.js (a server side Java Scripting platform), mainly targeting the Internet of Things space.

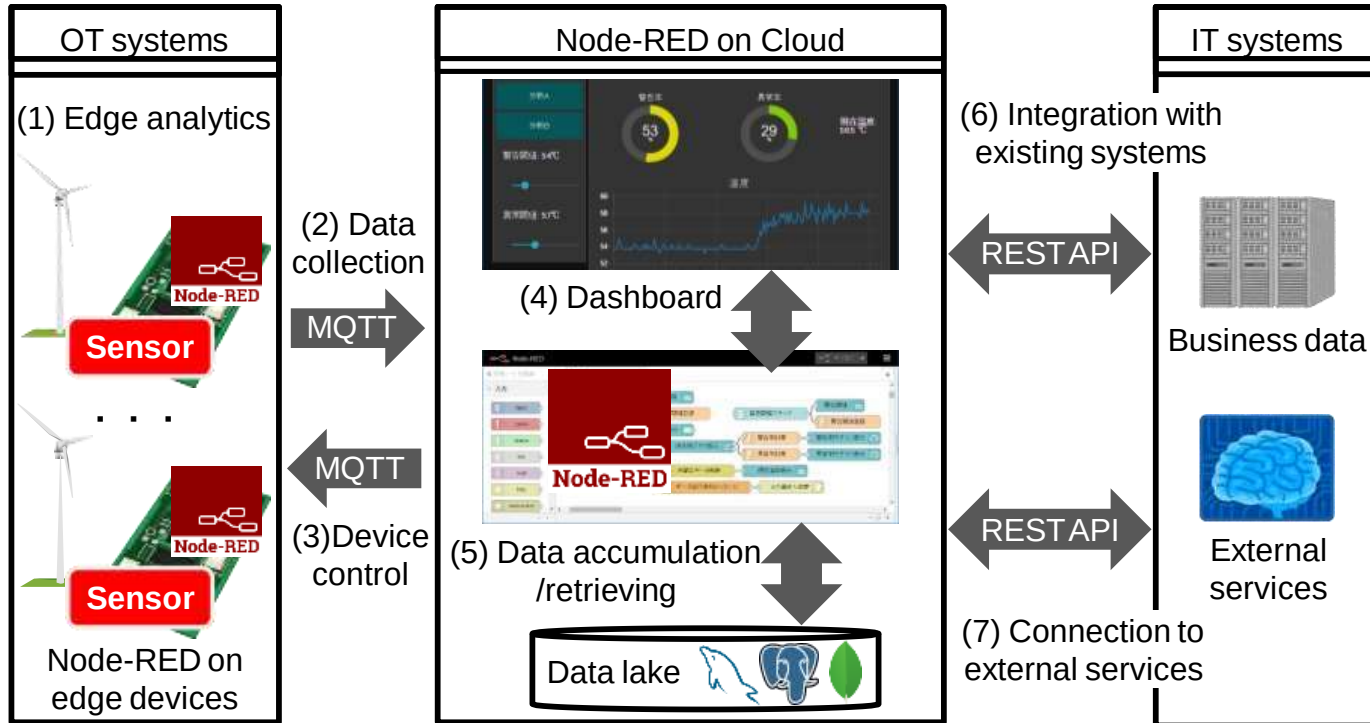


What is Node-RED?

- **Flow based** Programming which realizes quick development.
- Node-RED is a programming tool for wiring together hardware devices, APIs and online services in new and interesting ways.
- It provides a **browser-based editor** that makes it easy to wire together flows using the wide range of nodes in the palette that can be deployed to its runtime in a single-click.
- Various connectors(Node-RED nodes) to add functionalities
- Open source software under Linux foundation

What does Node-RED realize?

Node-RED has key features to realize IoT system.

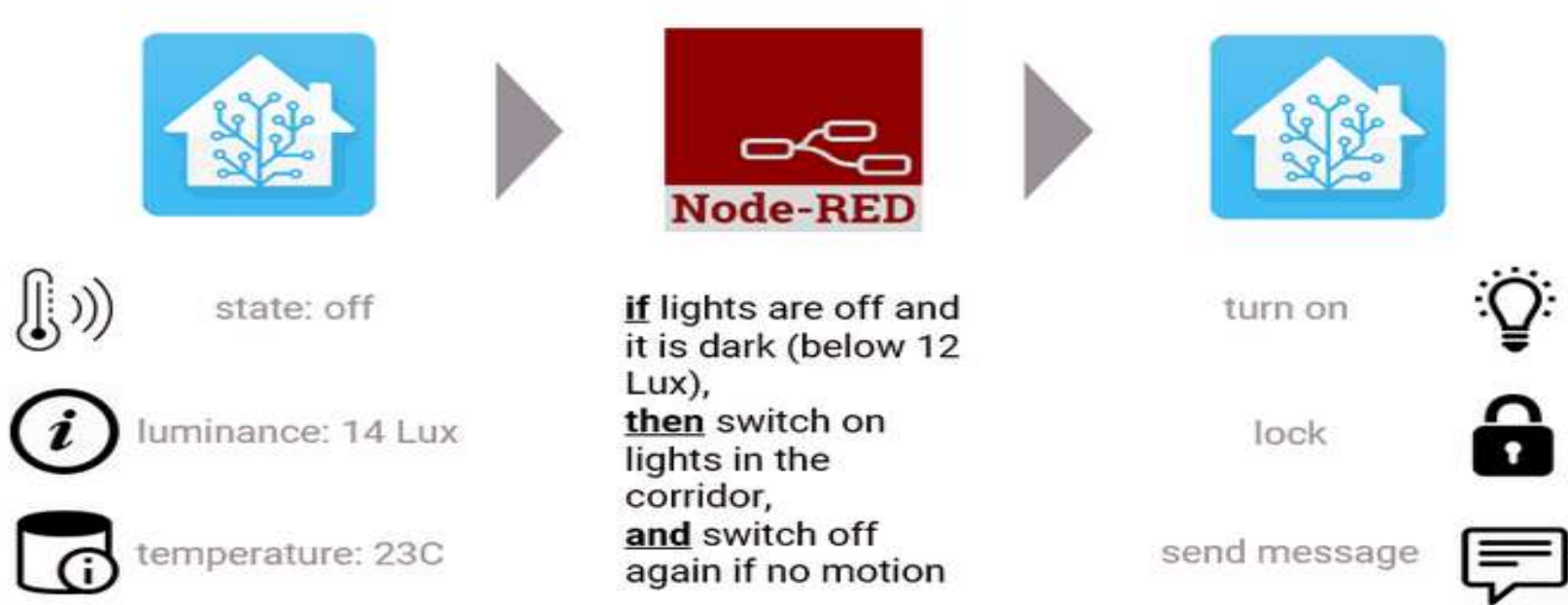


What is flow-based programming?

- **Flow-Based Programming (FBP)** is a slightly different approach to thinking about programming.
- The basic premise is to decompose the problem up into several components: data, processes, and the network.
- In FBP the applications are the collections processes, which linked using data Information Packets that travel between processes though defined connections.
- Compared to the usual way of thinking about the coding, flow-based programming is much more visual and you can almost draw the code.
- This makes it easy to learn and especially great for applications that need more mass adoption.



Architecture of interaction between Node-RED and Home Assistant



What is Node-RED?

Visual programming tools for IOT Applications

The screenshot displays the Node-RED web interface. On the left, the 'input' category is expanded in the 'filter nodes' panel, showing various nodes like inject, catch, status, link, mqtt, http, and webhooks. A red arrow points from the 'mqtt' node in this panel to a 'Sensor data via MQTT' node in the workspace, illustrating step (2). The workspace itself contains a flow: 'Sensor data via MQTT' (purple) connects to a 'temperature threshold' node (yellow), which then branches to 'Notification by e-mail' (green) and 'Visualize data on dashboard' (teal). Step (1) points to the 'mqtt' node in the panel, and step (3) points to the 'temperature threshold' node. On the right, a 'Deploy' button is highlighted with a callout for step (4). Below the workspace, a 'Node Help' panel for the 'mqtt in' node is visible, showing its description and output details.

(2) Drag and drop connectors to workspace

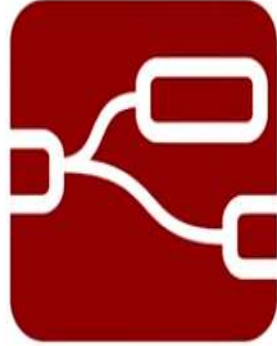
(4) Flow runs immediately once clicking deploy button

(1) Select connectors which has various functions

(3) Wire the connectors in the processing order



PROTOTYPING

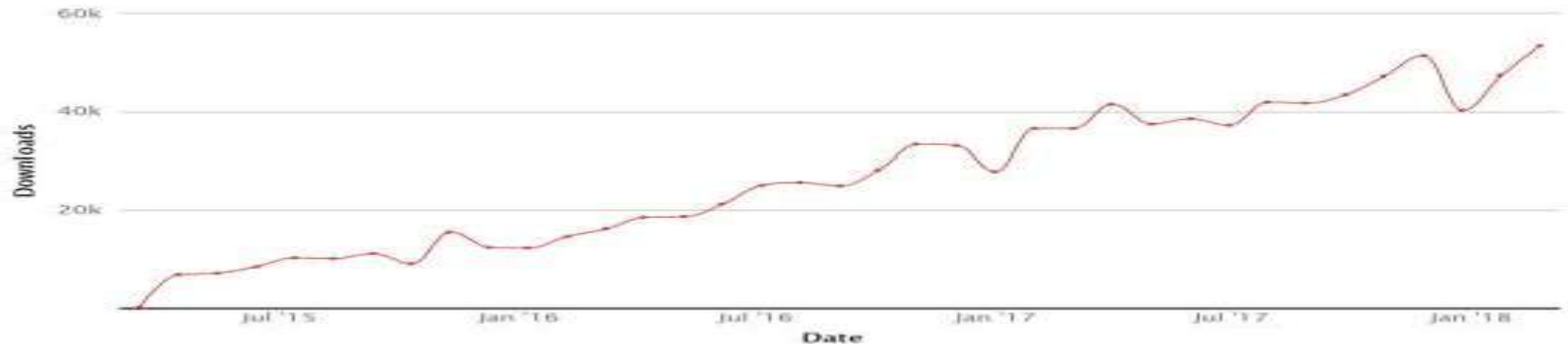


DEVELOPMENT

History of Node-RED

Node-RED has been more open and popular as open source software for both edge and cloud environments.

- March 2014: Released in QCon London.
- June 2014: Included in the catalog on IBM Cloud.
- November 2015: Pre-installed in Raspberry Pi.
- October 2016: Moved to Linux Foundation.
- As of March 2018: Downloaded 50,000 times a month.

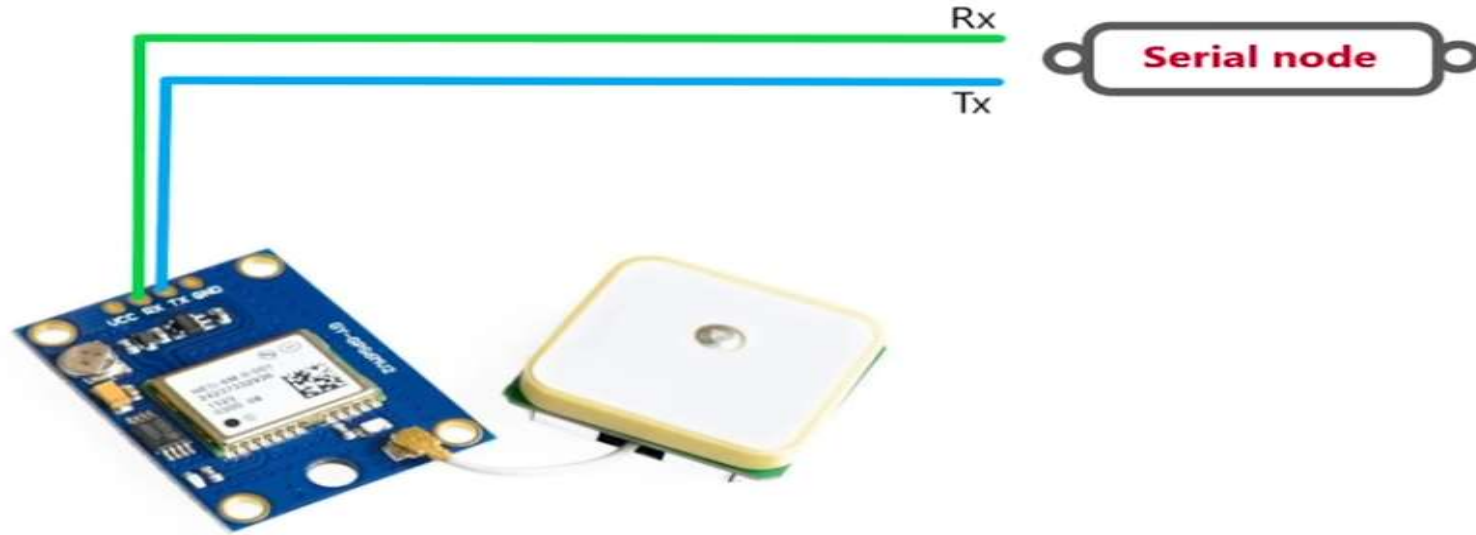


The number of download from npm repository

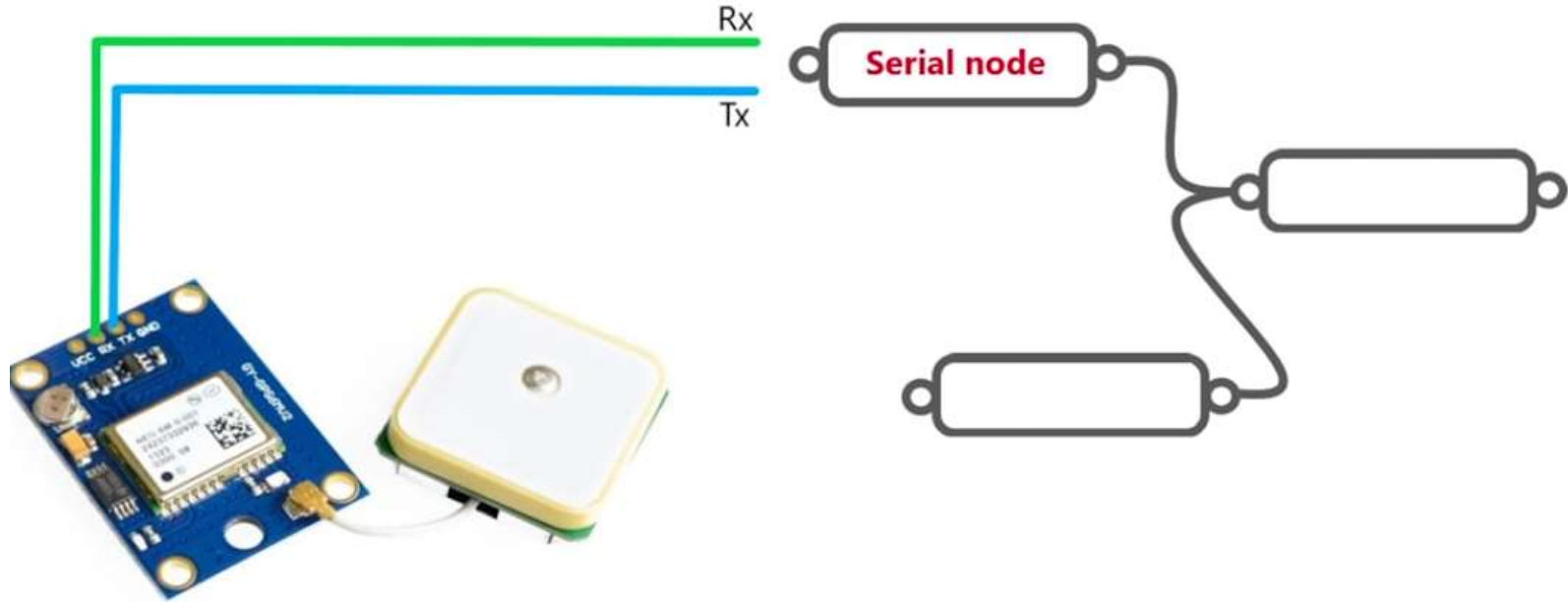
EXAMPLE



EXAMPLE



EXAMPLE



Why Node-RED?

(1) Rapid development for IoT applications

- IT and OT engineers can easily create IoT application within couple of hours without coding.
- Same development styles in both edge and cloud environments



(2) Standard technologies in industrial IoT

- Node-RED supports essential IoT technologies. (e.g. REST API, WebSocket and MQTT)
- Pre-installed software on standard edge devices

(3) Open community

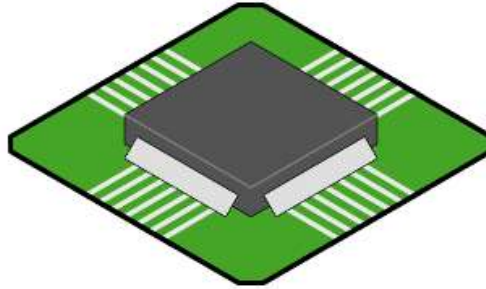
- 73 contributors including IBM, Sense Technic, and Hitachi are developing Node-RED on GitHub.
- 3rd party 1,345 connectors



Who can host Node-RED?

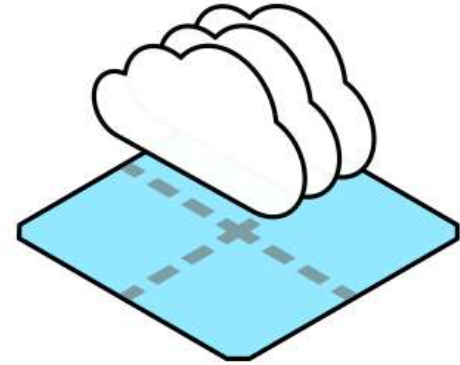


Run Locally



On a device

- Raspberry Pi
- BeagleBone Black
- Interacting with Arduino
- Android



In the cloud

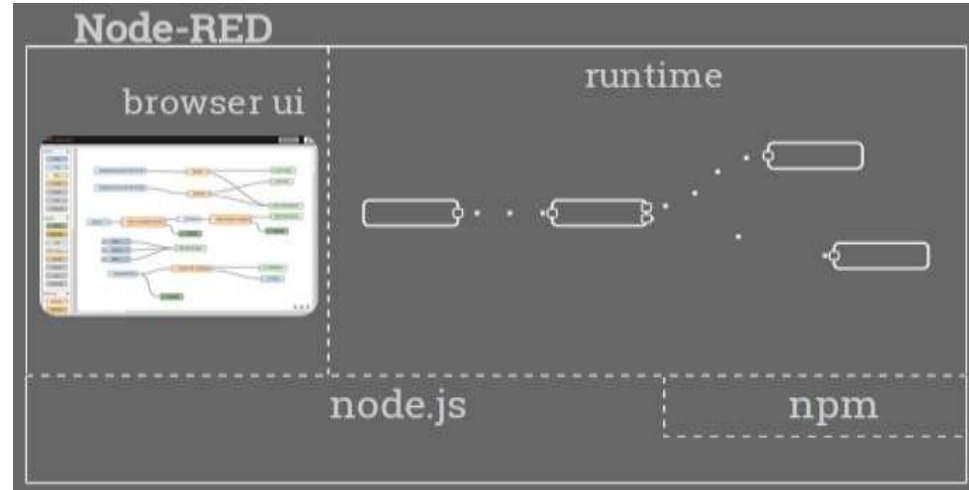
- IBM Cloud
- Amazon Web Services
- Microsoft Azure

Features of NodeRED



Architecture of Node-RED

- **Node.js v8-engine driven**; so it's fast
- **Event-driven, asynchronous io**; it's all about the events
- **Single-threaded event-queue**; built for fairness
- **Javascript front and back**; only one language runtime to deal with.



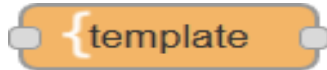
Basic Node types



- **Inject node**
 - Allows manual triggering of flows
 - Can inject events at scheduled intervals

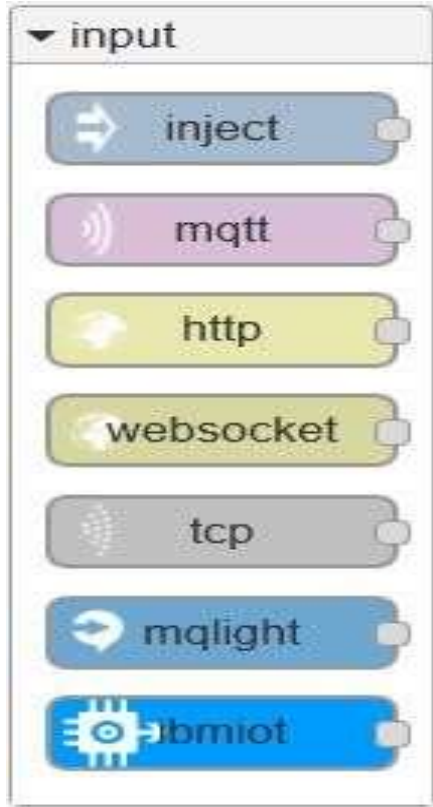


- **Debug node**
 - Show message content; either payload or entire object



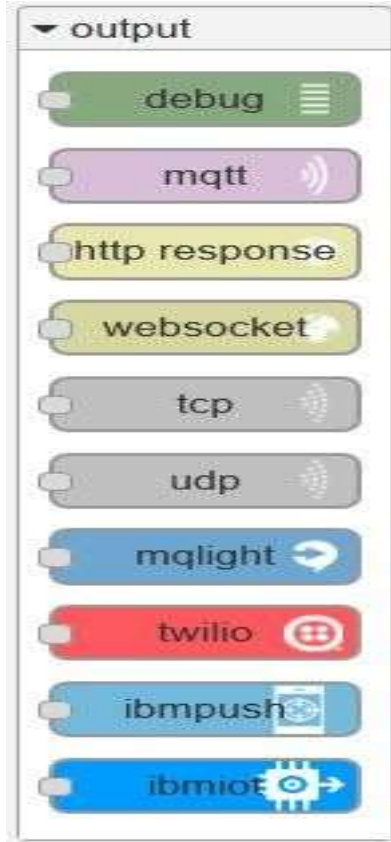
- **Template/ Function Node**
 - Modifies the output based on a Mustache Template

Input Nodes



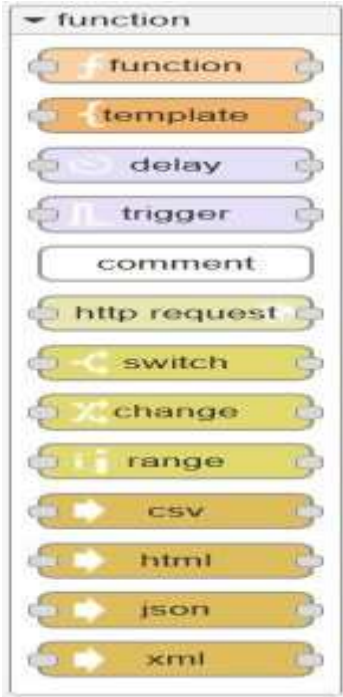
- **HTTP** – Act as an HTTP endpoint; great for building RESTful services
- **IBMIOT** – Receive messages from an attached IOT Foundation account
- Also can receive from Websockets, MQTT (pick your own broker), TCP and MQ Light

Output Nodes



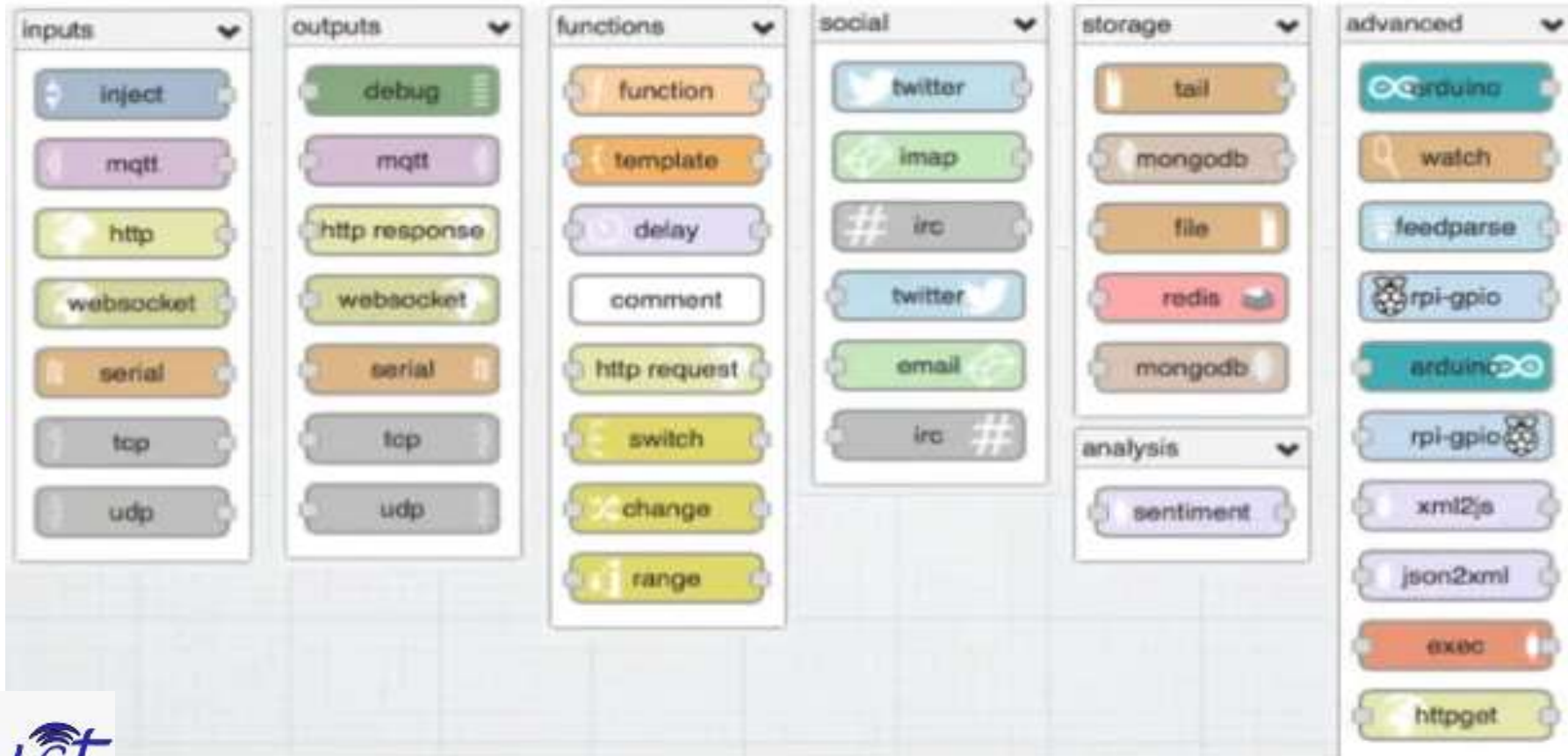
- HTTP Response; required as the final node when the input comes from an HTTP Request
- IBM IOT – send events out to the attached IOT Foundation account
- Twilio – send SMS messages via the Twilio service
- IBM Push – Send Push notifications to mobile devices
- Also can send requests through TCP, UDP, MQLight, WebSockets.

Function Node Types



- **Function node**
 - Run user-defined node.js code on the messages going by
 - Uses vm.createScript under the covers to sandbox execution
 - Console, util, Buffer included for convenience
- **Switch**
 - Change flow to different options based on a comparison

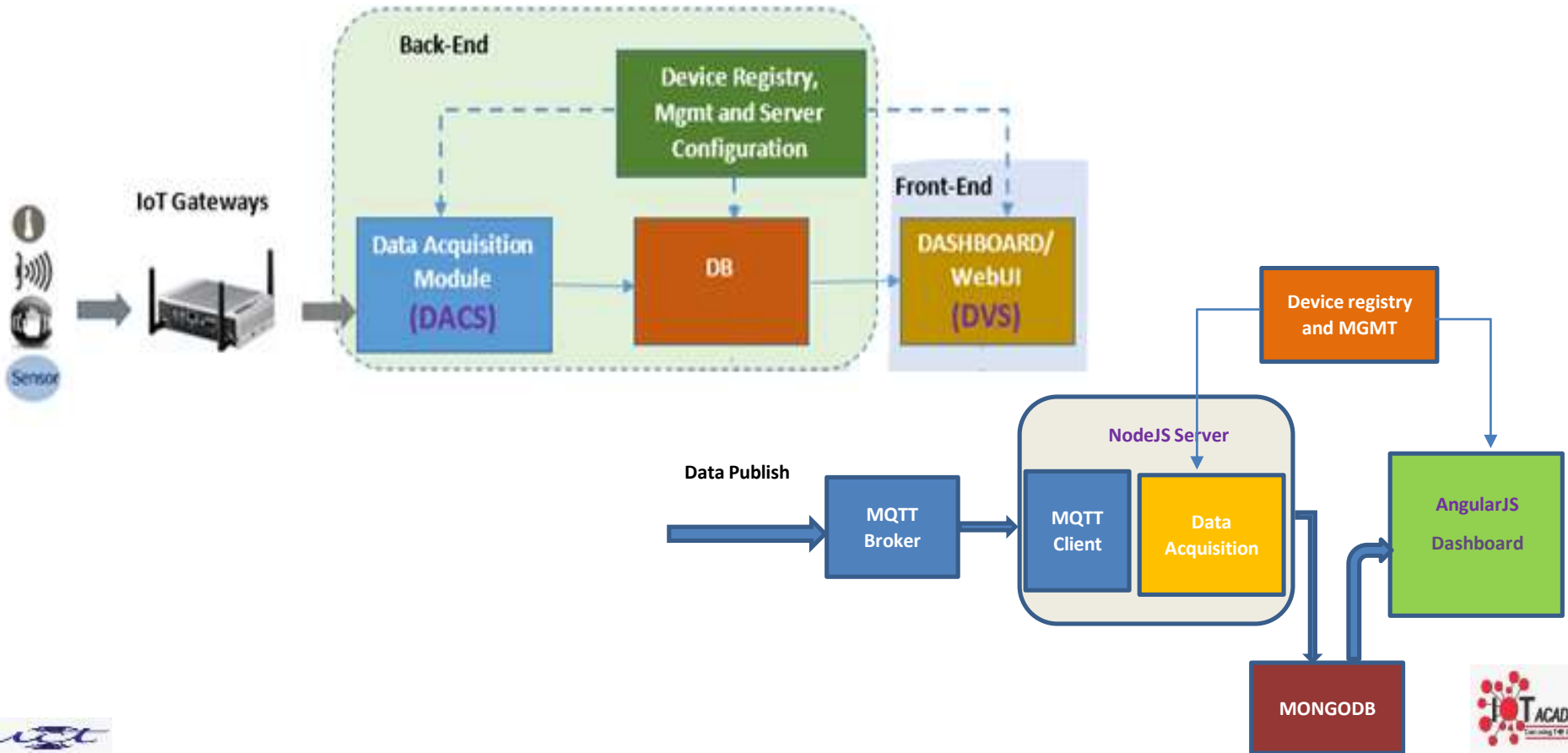
List of Nodes



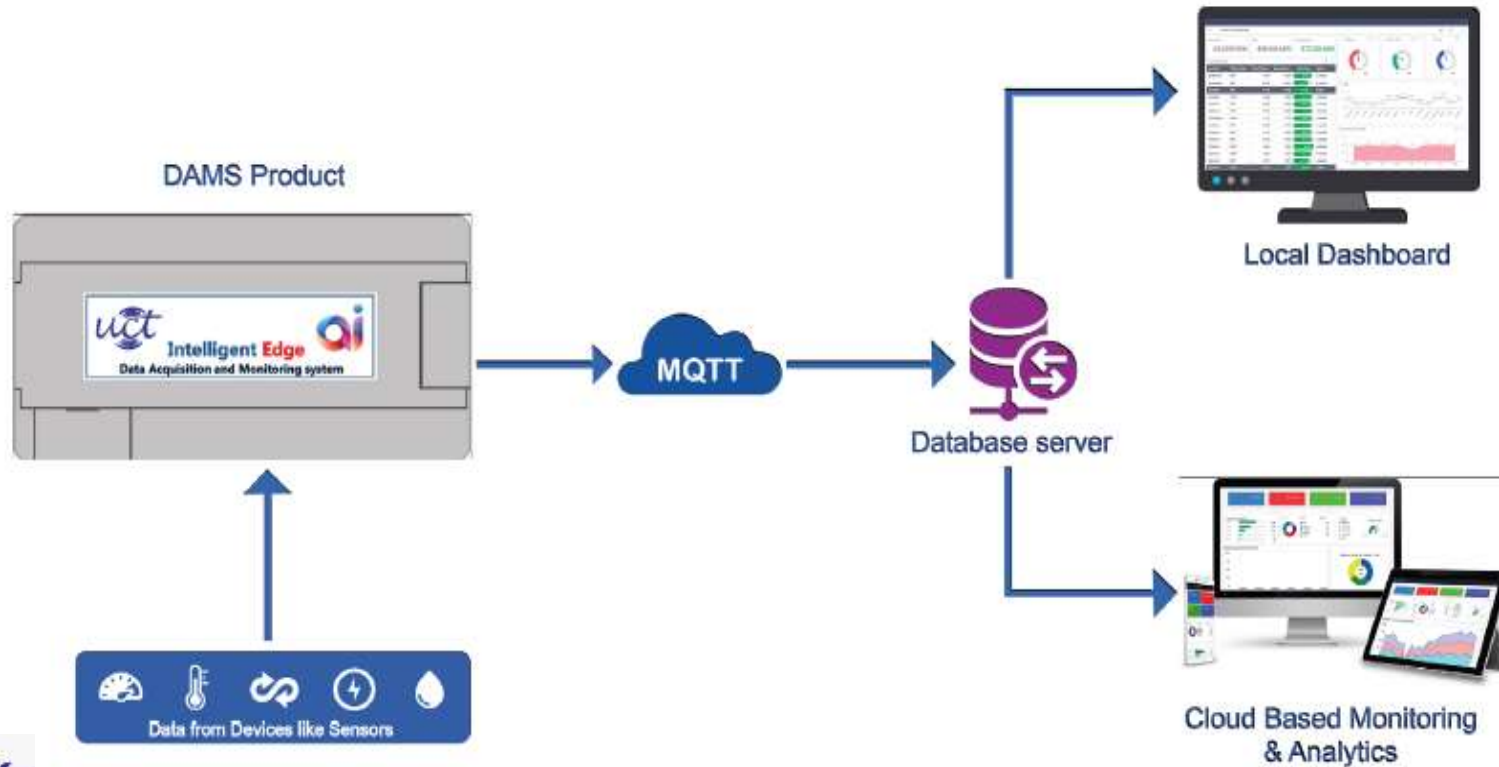
Need of a tool : NodeRED



IOT DATA FLOW IN ANY SOLUTION



Generic IOT Monitoring Solution



Approach to replace HWs with SW

Replace with Tinkercad

DAMS Product



Database server



Local Dashboard

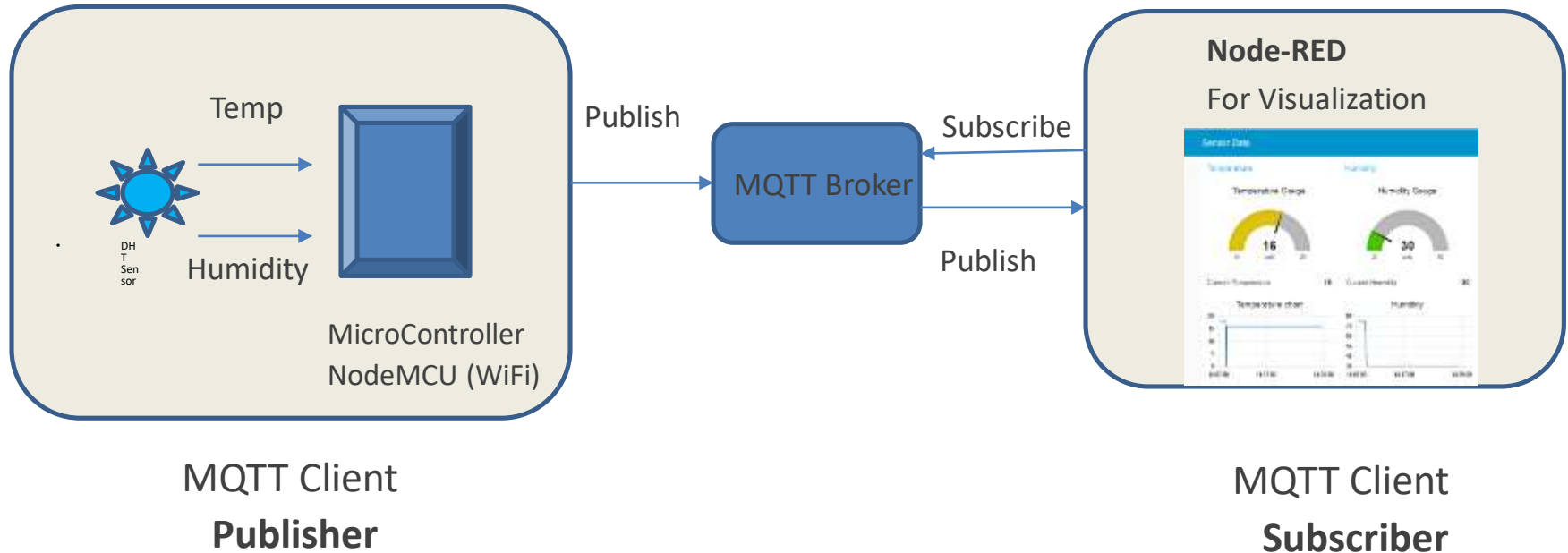


Design in Node-RED



Cloud Based Monitoring
& Analytics

Temperature Humidity Monitoring



Installation of NodeRED



Node-RED Installation Steps for Windows

Step 1. Search node js in Google and open downloads



node js



All



Books



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About 47,00,00,000 results (0.87 seconds)

nodejs.org ▾

Node.js

Node.js® is a JavaScript runtime built on Chrome's V8 JavaScript engine.

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Downloads. Latest LTS Version: 12.18.2 (includes npm 6.14.5). Download the **Node.js** source code or a pre-...

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Node.js

Node.js is an open-source, cross-platform, JavaScript runtime environment that executes JavaScript code outside a web browser.

[Wikipedia](#)

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Step 2.

(a) For Windows



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Node.js® is a JavaScript runtime built on [Chrome's V8 JavaScript engine](#).

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12.18.2 LTS

Recommended For Most Users

14.4.0 Current

Latest Features

[Other Downloads](#) | [Changelog](#) | [API Docs](#) [Other Downloads](#) | [Changelog](#) | [API Docs](#)

Step 2. (b) For Linux



Activities Firefox Web Browser

Node.js - Mozilla Firefox

Node.js

https://nodejs.org/en/

node JS

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Node.js® is a JavaScript runtime built on Chrome's V8 JavaScript engine.

New security releases to be made available June 2, 2020

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12.17.0 LTS Recommended For Most Users	14.3.0 Current Latest Features
--	--

Other Downloads | Changelog | API Docs Other Downloads | Changelog | API Docs

Or have a look at the Long Term Support (LTS) schedule.

Subscribe

Step 2. (c) For Mac



node.js

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12.13.1 LTS	13.1.0 Current
Recommended For Most Users	Latest Features

Other Downloads | Changelog | API Docs | Other Downloads | Changelog | API Docs

Or have a look at the [Long Term Support \(LTS\)](#) schedule.

Sign up for [Node.js Everywhere](#), the official Node.js Monthly Newsletter.

Step 3. Select Installer

LTS Recommended For Most Users	Current Latest Features	
 Windows Installer node-v12.18.2-x84.msi	 macOS Installer node-v12.18.2.pkg	 Source Code node-v12.18.2.tar.gz

Windows Installer (.msi)

Windows Binary (.zip)

macOS Installer (.pkg)

macOS Binary (.tar.gz)

Linux Binaries (x64)

Linux Binaries (ARM)

Source Code

32-bit	64-bit
32-bit	64-bit
	64-bit
	64-bit
	64-bit
ARMv7	ARMv8
node-v12.18.2.tar.gz	

Step 4. Open Command Prompt

Command Prompt

Microsoft Windows [Version 10.0.17763.1098]

(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\ [redacted] >



Step 5. write **node -v** command and check version of node js

Command Prompt

Microsoft Windows [Version 10.0.17763.1098]

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C:\Users\ [redacted] -node -v

 write this command

v12.18.1

C:\Users\ [redacted] _

 Version of node js

Step 6. write command **npm install -g node-red**

cmd Command Prompt

Microsoft Windows [Version 10.0.17763.1098]
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C:\Users\7...> npm install -g node-red_



Write this command

Wait for install

Step 7.write node-red

cmd Command Prompt - node-red

Microsoft Windows [Version 10.0.17763.1098]
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C:\Users\71421>node-red

Wait for some times

Step 8. Wait for this window and copy link at Browser

Select node-red

Microsoft Windows [Version 10.0.17763.1098]

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C:\Users\>node-red

30 Jun 20:31:47 - [info]

Welcome to Node-RED

=====

30 Jun 20:31:47 - [info] Node-RED version: v1.0.6

30 Jun 20:31:47 - [info] Node.js version: v12.18.1

30 Jun 20:31:47 - [info] Windows_NT 10.0.17763 x64 LE

30 Jun 20:31:51 - [info] Loading palette nodes

30 Jun 20:32:07 - [info] Dashboard version 2.15.3 started at /ui

30 Jun 20:32:07 - [info] Settings file : \Users\>\.node-red\settings.js

30 Jun 20:32:07 - [info] Context store : 'default' [module=memory]

30 Jun 20:32:07 - [info] User directory : \Users\>\.node-red

30 Jun 20:32:07 - [warn] Projects disabled : editorTheme.projects.enabled=false

30 Jun 20:32:07 - [info] Flows file : \Users\>\.node-red\flows_jitesh.json

30 Jun 20:32:07 - [warn]

Your flow credentials file is encrypted using a system-generated key.

If the system-generated key is lost for any reason, your credentials file will not be recoverable, you will have to delete it and re-enter your credentials.

You should set your own key using the 'credentialSecret' option in your settings file. Node-RED will then re-encrypt your credentials file using your chosen key the next time you deploy a change.

30 Jun 20:32:07 - [info] Starting flows

30 Jun 20:32:07 - [info] Started flows

30 Jun 20:32:07 - [info] Server now running at <http://127.0.0.1:1880/>

30 Jun 20:32:08 - [info] [mqtt-broker:1c35b1b.3a202ce] Connected to broker: mqtt://test.mosquitto.org:1883

Copy this link in Browser



filter nodes

Flow 1

Flow 4



common

inject

debug

complete

catch

status

link in

link out

comment

function

function

switch

change

info



Information

Flow "5e946e3c.ee1c18"

Name Flow 1

Status Enabled

Description

Hands-ON



Node-RED Hello World



- When you click on the Inject Node, it sends an event through the flow – triggering the template node and sending the result to the Debug node

Other Hands-ON

- Please check all exercises at
- <https://cookbook.nodered.org/basic/>
- Convert timestamp to array hours, minutes, day, month, year
- <https://flows.nodered.org/flow/ba071f83d91d2b065074>
- Export the flows copied from above exercises to your NodeRED browser
-

Study

<https://nodered.org/docs/user-guide/writing-functions>

<https://nodered.org/docs/user-guide/nodes>

